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Service Design Perspectives on Social Interaction with Robots in Kiosk-Style Encounters

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Figure 1: Furhat robot in a service situation during the user study in a restaurant.

Abstract

Service robots are increasingly entering public service environments, yet their acceptance and role suitability in brief service encounters remain unclear. This paper examines humanoid robots in restaurant contexts through an in-the-wild deployment of the Furhat robot (n = 23), investigating real-time usability, perceived competence, discomfort, and affective responses. The field study showed that curiosity and amusement were common, but often accompanied by discomfort and a preference for human service in core encounters. Evaluations of usability and competence were shaped by first impressions, expressivity, and responsiveness during the short interaction. Building on these findings, we propose a service design oriented framework, visualized as a service blueprint, to support the analysis and design of robot-mediated kiosk-style encounters. Our work highlights the value of service design perspectives for human–robot interaction research.

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CCS Concepts

• **Human-centered computing** → **Interactive systems and tools**; **Interaction design**; **HCI design and evaluation methods**; **User centered design**.

Keywords

Service robots; Human–robot interaction; Service design; Perceived competence; User experience; Restaurant services

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1 Introduction

Artificial intelligence and robotics are reshaping everyday services across healthcare, hospitality, entertainment, and education [6, 29, 32]. As systems become more capable and affordable [31], service providers have started to deploy robots to handle routine customer-facing work [12]. Restaurants are a clear example: demand fluctuates, queues form quickly, and staffing can be difficult during peak hours. In these conditions, a robot can be attractive

as a front-of-house interface that supports order intake, provides consistent messaging, and reduces pressure on staff, yet its presence may also change what customers feel is acceptable at the service counter and how the encounter is experienced [18].

Restaurant deployments have often been discussed at the level of roles and process integration, e.g., through service blueprinting [17] and early catering deployments using platforms such as the Pepper robot [23]. Still, less is known about very brief, transactional encounters where a customer approaches a service point, completes a bounded task, and leaves. In such moments, evaluations are formed quickly and can be shaped by social cues, breakdown handling, and whether the interaction feels smooth enough to finish the transaction. This paper addresses that gap by examining a short order-taking encounter delivered by a humanoid robot in a live restaurant.

A humanoid robot differs from a conventional kiosk because it can utilize conversational timing, attention cues, and expressive behavior as part of the service interface. Prior work in service research and human-robot interaction suggests that these cues can affect first impressions, comfort, and expectations about whether the system will perform the task correctly [30]. In a short transaction, these constructs are closely related in practice: (1) customers' self-reported affective responses immediately after the encounter, (2) perceived competence and discomfort as judgments of the robot as a service agent, and (3) perceived usability as a proxy for whether the interaction was easy to complete under real conditions [5, 19]. We treat these as early-stage acceptance signals in a brief service touchpoint.

We study these signals using the Furhat robot [11], a head-only socially interactive robot with a rear-projected face and movable head. Furhat can produce facial expressions, synchronized lip movements, and directed gaze, which supports controlled design of attention cues and confirmations during conversation. This matters in transactional encounters, where the customer needs clear turn-taking and confirmation to proceed. We deployed Furhat as an ice cream order-taking interface in an in-the-wild field study in a restaurant (n = 23) (Figure 1). We selected an in-the-wild setting to capture the conditions that shape short public encounters, such as bystanders, ambient noise, queue pressure, and group participation, which are difficult to reproduce in a lab.

We use the term *kiosk-style encounter* to mean a brief, stand-alone touchpoint where customers initiate the interaction, complete a bounded task (e.g., ordering and paying), and leave without an ongoing relationship. Within this scope, *transactional trust* refers to the expectation that the agent will capture the order correctly, behave predictably and appropriately, and support smooth completion of the transaction. In this paper, we examine trust-related expectations in relation to perceived competence and discomfort, along with qualitative accounts of concerns raised by participants.

This paper addresses the following research questions:

- **RQ1:** What affective responses do customers report after a brief humanoid robot encounter in a restaurant setting?
- **RQ2:** How do these affective responses relate to perceived competence, discomfort, and perceived usability in the same encounter?

Our findings indicate that participants were often curious and amused by the robot, yet many still preferred a human for core service moments. Robots were viewed as suitable for limited or supporting tasks, and participants' evaluations were sensitive to first impressions and the robot's expressive behavior during the short interaction.

We contribute to research on service robots and human-robot interaction by: (1) reporting results from an in-the-wild deployment examining customer reactions to a humanoid robot in a restaurant; (2) characterizing early-stage acceptance signals in brief transactional encounters, combining affective responses with perceived competence, discomfort, and usability; and (3) proposing a service design-oriented model, expressed as a service blueprint for robot-mediated kiosk-style encounters, that links touchpoints to observed breakdowns and customer responses.

2 Related Work

Service robots have been studied from the perspectives of human–robot interaction, technology acceptance, and service design. To position our work, we review prior research on service encounters, acceptance, trust and social–emotional interaction, and service design approaches. We contrast prior work focusing on extended interactions with the context of brief, transactional kiosk-style encounters.

2.1 Service Robots in Service Encounters

Wirtz et al. [30] define service robots as autonomous and adaptable interfaces that engage, communicate, and deliver services. Such robots may provide services directly to customers or indirectly through supportive systems [20]. Unlike industrial robots operating in controlled environments, service robots operate in dynamic, socially demanding contexts where short, transactional encounters are integrated into customer journeys, such as kiosks, check-in desks, or shopping assistance [22].

HRI research has examined how robots can occupy front-stage roles, i.e., highly visible touchpoints that shape customer experience [1]. In service settings such as hotels, airports, and restaurants, robots have been deployed as receptionists, information providers, and transaction handlers [21]. In such settings, the robot's performance and social presence are critical to the overall perceived service quality. Carpinella et al. [5] reported that embodiment, interactivity, and adaptability strongly influence customer perceptions of service robots. In kiosk-like contexts, robots may combine the efficiency of self-service technologies with the social presence of human-like interaction, creating a hybrid service experience [22]. Early work examined how Pepper could be introduced into catering services roles such as welcoming, order taking, and reception [17]. Because public settings are unpredictable, robots must manage diverse expectations and communication styles, making service encounters a challenging domain for design.

Together, this work shows how service robots can occupy visible front-stage roles and shape customer experience, yet it remains unclear how customers evaluate robots during very short, kiosk-style encounters where acceptance decisions are formed quickly.

2.2 Acceptance of Service Robots

Acceptance has been examined through frameworks such as the Technology Acceptance Model (TAM), the Unified Theory of Acceptance and Use of Technology (UTAUT), and the Almere model [13]. These identify perceived usefulness, ease of use, enjoyment, and social presence as key parameters. In service encounters, enjoyment and social presence are especially important, as customers expect both efficiency and meaningful interaction [8]. Anthropomorphic features, such as humanlike appearance or speech, can enhance engagement and increase acceptance [9, 15]. However, excessive anthropomorphism may raise unrealistic expectations that reduce acceptance [7]. Sharma et al. [27] found that perceived enjoyment, speed, and novelty drive customer satisfaction with robotic service in restaurants, while perceived usefulness did not; satisfaction and trust mediated the intention for repeat usage. Recent work has further extended TAM to restaurant settings with robot waiters, showing how established acceptance models continue to evolve in response to service robot adoption [10]. Overall, research indicates that acceptance results from the interplay of technology performance, anthropomorphic design, and the service context.

While acceptance models identify key factors such as usefulness, ease of use, enjoyment, and anthropomorphism, most studies focus on intention to use or repeated interaction, leaving open how acceptance is formed in brief, one-off service encounters.

2.3 Trust and Social-Emotional Interaction with Service Robots

Trust is central in service contexts where users depend on robots to provide information or complete transactions. Trust is typically described through competence, reliability, predictability, and benevolence [25]. Kim et al. [15] identified that personalization can enhance trust, as personality expressed through appearance and behavior can support social interaction, and strengthen human–robot connection. Expressive cues such as gaze, facial movement, and tone of voice contribute to the robot's personality and can increase trust [3]. Kirjavainen et al. [16] reported that appropriate robot facial expressions supported pragmatic aspects of the interaction, whereas inappropriate expressions could be confusing yet still increase user engagement. In short service encounters, trust must be established almost immediately. Expressive behavior, politeness, and responsiveness are crucial for positive first impressions [24], while mismatches between personality and actual capability can rapidly undermine trust. However, much of this prior work assumes sufficient interaction time for trust to develop, whereas in short public service encounters, trust-related judgments must be formed almost immediately.

2.4 Service Design Perspectives on Human–Robot Interaction

Service design provides a lens for understanding robots as actors in service systems. Concepts such as service encounters, customer journeys, and touchpoints emphasize that value is co-created between provider and customer [1]. When robots are introduced, they become visible parts of the interface and can reshape the overall experience.

Several HRI studies have adopted service design perspectives. Mutlu and Forlizzi [22] highlighted how workflow and organizational factors shape robot roles, while Eyssel et al. [9] showed how expectations of anthropomorphism affect how users perceive frontline robots. Later work has considered robots as co-performers in encounters, influencing perceptions of quality, efficiency, and fairness [30].

Service design approaches stress prototyping and co-design in developing robotic services. Experience prototyping allows stakeholders to explore how robots alter customer journeys and where breakdowns may occur [28]. This positions robots as one element within a broader system of people, environments, and technologies. In kiosk-type applications, a service design view highlights how trust, acceptance, and social-emotional interaction contribute to the perceived service quality. Closest to our work, Lai and Tsai [17] used service blueprinting to map Pepper's roles in catering services, which we extend by examining how the more expressive Furhat can shape trust, social-emotional interaction, and acceptance in brief encounters. Despite these contributions, service design methods have rarely been used to empirically connect observed affective responses and breakdowns in short robot encounters to specific service touchpoints.

2.5 Gaps and Positioning

Prior research has examined service robot roles, acceptance, and trust, often from a service design perspective that treats robots as actors embedded in customer journeys. However, much HRI research emphasizes long-term or repeated interactions in domestic or care contexts [8, 13], while brief, transactional encounters such as kiosks or restaurant counters remain underexplored [17]. In these settings, first impressions and early breakdowns strongly shape acceptance.

The Furhat robot used in our study provides an embodied platform for supporting trust through nuanced expressions, gaze, and a speech interface that enables conversational grounding. Prior studies have reported that Furhat is perceived as more trustworthy than less expressive robot platforms such as Pepper, owing to its ability to convey subtle social cues [14]. However, little work has examined how Furhat or similar expressive robots build acceptance and trust in kiosk-style settings, where efficiency must be balanced with social presence. We address this gap through an in-the-wild user study that situates Furhat as an ice cream kiosk vendor, allowing us to investigate acceptance, trust, and social-emotional interaction under real service conditions.

3 Field Study in a Restaurant Context

We conducted an in-the-wild study to examine whether an expressive robotic platform such as Furhat could address challenges reported in prior HRI studies of public-facing service robots, including discomfort, limited perceived competence, and weak social presence during brief encounters. Situating Furhat as an ice cream kiosk vendor, the study explored how its facial animation, gaze coordination, and conversational behavior influence perceived competence, acceptance, and customer experience in an authentic service context. We investigated how Furhat's controlled gaze and facial animation



Figure 2: The study setting at the restaurant. Right: Setting from the robot's viewpoint.

might affect users' emotional reactions, competence-related expectations, and perceived competence during short service encounters in a real restaurant environment.

An in-the-wild deployment was chosen to capture conditions that shape brief public encounters, such as bystanders, ambient noise, time pressure, and group participation, which are difficult to reproduce in controlled laboratory studies but directly affect first impressions and breakdown handling.

3.1 Study and Prototype Design

Order taking was identified as the most acceptable role for a humanoid robot based on prior restaurant robot studies and service design work that highlight clearly bounded, low-risk front-stage tasks as suitable entry points for robotic services [17]. Hence, we designed a service concept prototype called *Robot Ice-Cream*, featuring the Furhat robot as a kiosk-style salesperson. The study took place in a local restaurant in [anonymized] city center, which had not previously used robots in customer encounters. We chose to use the Furhat humanoid robot as our robotic platform for our in-the-wild field study. Furhat differs from humanoid platforms that have been extensively used in prior HCI research, such as the Pepper robot [23], by offering fine-grained facial animation and eye-gaze coordination that enable controlled variations in expressiveness. This made it suitable for examining how subtle expressive cues influence user perceptions in kiosk-type encounters.

To stage the prototype, we constructed a stylised ice-cream kiosk that integrated with the restaurant's servicescape. The kiosk was designed to reflect the venue's visual identity and to support smooth interactions for customers of varying heights. Clear signage signaled the robot's role as an ice-cream vendor. The stand was built using lightweight materials (cardboard, paint, tape), reflecting a low-fidelity prototyping approach typical in service design.

For the robot's role, we created the personality profile *Ilona, the Ice-Cream Vendor*. Using ChatGPT to generate an initial sketch, we refined the traits to fit the restaurant's values and service context. The final profile was imported into Furhat Studio's AI Creator, a no-code tool for configuring dialogue, facial expressions, and emotional delivery [11]. We selected a predefined face *Isabel* and voice *Suvi-Neutral*. To initiate encounters, the robot greeted customers with the phrase: "Hello! I am Ilona, welcome to the ice cream bar. How can I help you?" Subsequent dialogue was AI-generated in real time, based on the provided personality description and contextual information such as menu items and flavors.

The kiosk was positioned opposite the restaurant's cash register to create a visible, accessible touchpoint while keeping the interaction distinct from the normal payment flow. The study ran for three hours during the restaurant's normal lunchtime service. Figure 2 shows the user study setting.

3.2 Study Procedure

Normal restaurant customers were invited to participate in the study and interact with the robot during their visit. Participants received an ice cream as compensation for their involvement. Before starting the study, participants were informed about the study and provided informed consent. Each interaction began when the participant greeted the robot. The robot responded with its scripted introduction and guided the customer through two tasks: selecting an ice-cream flavor and choosing a payment method.

Following the interaction, participants completed a survey that included background questions and standardized scales. The Robotic Social Attributes Scale (RoSAS) [5] was used to measure perceived warmth, competence, and discomfort, and the UMUX-Lite [19] to evaluate usability. In addition, participants indicated their preferred type of service agent in a restaurant (robot, self-checkout, or human) and described the emotions evoked by the robot. The study moderators took field notes during interactions, focusing on observable behaviors such as hesitation, breakdowns, and bystander involvement.

A reflexive thematic analysis following Braun and Clarke [2] was conducted on open-ended responses, with themes discussed and refined collaboratively by the author team. UMUX-Lite scores were converted to approximate System Usability Scale (SUS) values [4] to allow comparison with prior datasets.

Study Participants. Altogether, 23 participants took part in the study (n = 23). Participants' ages ranged from 19 to 65+ years (M = 38): three were 19–25, seven were 26–35, six were 36–45, two were 46–55, two were 56–65, and one was 65+. Fifteen participants self-identified as men (62.5%) and eight as women (37.5%). Prior familiarity with robots varied: robot vacuum cleaners (52%), robotic pets (39%), and lawnmowers (26%) were the most common. Two participants reported having no prior experience with robots.

3.3 Results

The results combine participants' questionnaire responses and thematic analysis of open-ended feedback with moderators' observations. Together, these findings highlight how customers first engaged with the robot in the restaurant servicescape, how they assessed its usability and social attributes, and what emotions were evoked. The following subsections present these results in terms of participants' initial engagement, usability assessments, perceived social attributes, and emotional responses. Together, these results address RQ1 by characterizing reported affective responses and RQ2 by examining how these responses co-occur with perceived competence, discomfort, and usability.

Initial impressions and interaction in the servicescape. Conducting the study in a live restaurant enabled spontaneous participation from a wide range of customers, potentially including individuals who would not normally take part in robotics research. This exposed diverse, unexpected reactions that are difficult to capture in controlled laboratory settings. The Furhat robot attracted attention in the restaurant, with many customers approaching out of curiosity or commenting that they found the robot humorous. Some observed the interaction from their tables before engaging, and at times queues formed around the kiosk, partly driven by the ice cream compensation. Group interactions were common, as participants often approached with friends, reflecting the social dynamics of the restaurant setting. Nervousness was sometimes evident at the start, with a few customers seeking reassurance from the moderator, while others initiated interactions independently. Once engaged, participants generally became more confident, with several choosing to continue interacting beyond the minimum task.

Usability and preferred service agents. Usability ratings from the UMUX-Lite responses were converted to SUS scores, resulting in a mean score of 60 (SD = 0.9), indicating a moderately positive user experience. When asked to choose a preferred service agent in a restaurant, the majority of participants (70%) selected a human waiter, while 22% preferred self-service checkouts. These results indicate that participants generally preferred human service for core encounters, while viewing the robot as a novel or supplementary touchpoint.

Social attributes of the robot. Responses to the Robotic Social Attributes Scale (RoSAS, 1–5) are presented in Figure 3. These produced mean scores of *Warmth* = 3.1 (SD = 1.1), *Competence* = 3.1 (SD = 0.8), and *Discomfort* = 1.9 (SD = 1.2). Participants thus perceived the robot as somewhat warm and moderately competent, with relatively low discomfort. However, there was considerable variation between individuals. Pearson correlation analysis revealed a moderate positive relationship between usability and competence, $r(21) = .47$, $p = .025$, indicating that participants who rated the robot as more usable also tended to perceive it as more competent. Correlations with warmth and discomfort were weak. This pattern suggests that perceptions of interaction smoothness were associated with judgments of competence, providing a link between participants' experiential responses and their evaluation of the robot as a service agent.

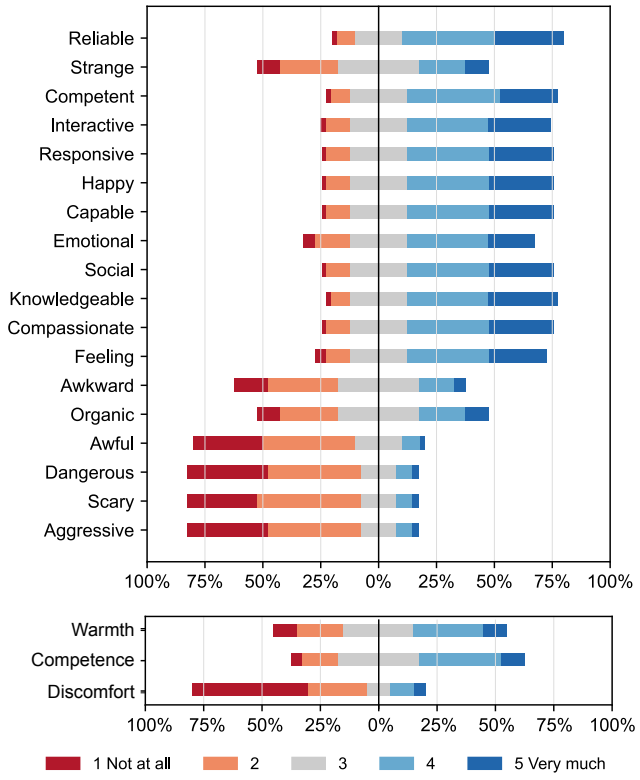


Figure 3: Top: Ratings for individual RoSAS variables. Bottom: Mean ratings for the RoSAS dimensions.

Emotional responses. Open-ended survey responses about the emotions evoked by the robot were coded into five themes: *Positive emotions*, *Conflicted emotions*, *Negative emotions*, *Critical*, and *Future-oriented*. The most frequent category was *Conflicted emotions* (8 comments, 42%), followed by *Positive emotions* (6, 32%), *Future-oriented* (3, 16%), and both *Negative emotions* and *Critical* (2 each, 11%). These themes were interpretively mapped to RoSAS dimensions based on the content of participants' comments (Table 1). For example, *Positive* and *Future-oriented* comments aligned with perceptions of warmth, while *Critical* remarks related to competence, and *Negative* and *Conflicted* comments reflected discomfort. Together, these responses highlight how participants balanced curiosity, amusement, and optimism about the future with concerns about awkwardness, reliability, and situational fit.

Theme	Example quotes	Mapped RoSAS dimension(s)
Positive emotions	"It told stories on its own initiative, which was nice" (P14); "It was fun, engaging, and sympathetic. It evoked joyful and fun emotions" (P17); "The robot aroused interest" (P14)	Warmth
Conflicted emotions	"A little conflicting. Would work with some applications" (P13); "Awkwardness, interest and thankfulness" (P7)	Warmth, Discomfort
Negative emotions	"A bit awkward" (P5); "Strange" (P18); "It was disturbing that the robot's eyes sometimes looked away when we were talking" (P19)	Discomfort
Critical	"If the robot worked flawlessly, I could use it, but its slow processing raises concerns about how well it reacts and works" (P6); "Too chatty" (P24); "The loud talking in the background disrupted the robot's hearing" (P14)	Competence
Future-oriented	"Great opportunities, even though it still needs some developing" (P3); "Necessary in the future" (P20); "Jobs will also be needed in the future. Maybe in some jobs robots would be OK too" (P23)	Competence, Warmth

Table 1: Identified themes in participants' emotional responses to the customer service robot, with representative quotes and corresponding RoSAS dimensions. Each participant contributed at most one coded emotional response. Theme frequencies are reported in the accompanying text.

4 Discussion

This study investigated how reported affective responses and user experience influence acceptance and perceived competence of a humanoid robot in a restaurant setting. Taken together, these findings address RQ1 by characterizing reported affective responses and RQ2 by showing how these responses align with perceived competence, discomfort, and usability in brief service encounters. Building on these results and prior work, we discuss how first impressions, interaction flow, and servicescape conditions shape short service encounters, and how these insights inform a service design oriented interpretation of robot-mediated kiosk encounters.

4.1 First Impressions and Interaction Flow in Short Encounters

Our findings show that first impressions strongly shaped the direction of the service encounter. Small timing differences in gaze, greeting length, and turn-taking influence whether the robot felt warm or awkward, supporting prior HRI work showing that social cues shape first impressions and competence-related judgments in human–robot interaction [24, 25].

Many customers watched others first, then approached once the path to action was clear. Despite open conversation, most interactions settled into a familiar structure of greeting, flavor choice, and payment, which reduced uncertainty. Common breakpoints were: hearing difficulties in noise, extended small talk that some enjoyed and others found tiring, and missing exit cues that left people unsure when to leave. As one participant noted, "The robot aroused interest… the loud talking in the background disrupted its hearing, but it still told stories on its own initiative, which was nice" (P14).

Furhat's back-projected face, controllable gaze, and short confirmations enabled quick acknowledgments [14], but reactions still varied, as reported by prior work on anthropomorphic design [9]. Participants often highlighted the importance of clear entry and exit cues, which provided structure, supported smoother conversation flow, and helped prevent overly long turns.

4.2 Positioning Robots within Service Roles and Contexts

Prior analyses of workflow and organizational integration [22] and frontline service research [21, 30] have reported a preference for human staff in core encounters such as ordering, with robots more often positioned in supporting or peripheral roles. Earlier work utilizing the less expressive Pepper robot highlighted welcoming, reception, and booking confirmation as initial roles in restaurant services [17]. Our study extends this with a more expressive robot head, showing that richer greetings and confirmations are possible, yet role clarity and servicescape design remain decisive [1].

4.3 Design Implications Mapped through a Service Blueprint

We use service blueprinting to translate the empirical findings into actionable design insights by summarizing journey stages, observed frictions, design moves, trust cues, and measures (Table 2). Figure 4 visualizes the "Robot Ice Cream" encounter as a service blueprint mapping the customer journey across frontstage interactions, backstage processes, and physical evidence. It shows how participants approached the kiosk, interacted through greeting, ordering, and payment, and then reflected on the experience, while also making visible backstage processes such as speech recognition and staff fallback protocols. Mapping emotional dynamics onto these layers highlights key breakdowns—uncertainty at entry, difficulties ending conversations, and hearing impairments in noisy environments—and points to concrete opportunities for design.

The blueprint links these observations with validated scales: RoSAS indicated moderate Warmth and Competence with low Discomfort, and perceived competence was correlated with usability, echoing prior findings on performance and acceptance

Journey stage	Observed friction	Design move	Trust cue	Measure
Approach	Uncertainty about what to do; bystanders observing before engaging	Clear role signage and short loop that states capabilities	Friendly head turn, nod, one-line offer	Approach rate; drop-off
Greeting	Uneven first seconds; gaze felt odd to some	Fixed-length greeting with quick handoff	Polite acknowledgment, backchannels	RoSAS Warmth; first-turn time
Order	Mishearing in noise; long answers	Confirm each slot; offer touch fallback	Short confirmations; visible progress	Task success; UMUX-Lite
Payment	Low tolerance for delay	Pre-announce steps; progress indicator	Clear confirmation of amount and outcome	Time on task; errors
Exit	No clear end signal; lingering	Closing line plus visual return to idle	Thank you and subtle gaze away	Queue flow; repeat approach

Table 2: Service blueprint summary linking journey stages, frictions, design moves, trust cues, and measures.

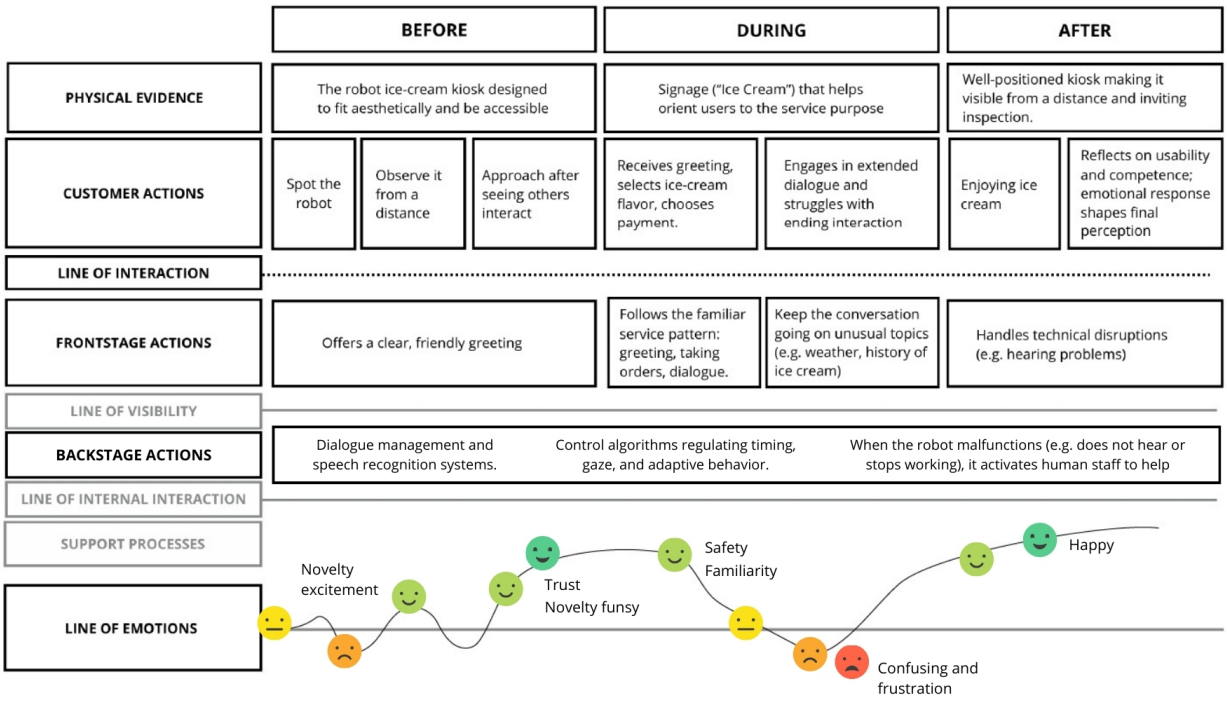


Figure 4: Service blueprint of the "Robot Ice Cream" encounter, highlighting emotional dynamics across the customer journey. The focus is on frontstage interactions between customers and the robot.

[10, 25, 27]. By combining RoSAS [5] and UMUX-Lite [19] with service blueprinting, the blueprint serves as a reusable analysis aid that helps designers identify breakdowns and competence-related concerns at specific service touchpoints and support coordinated iteration in the service context [26].

4.4 From Functional to Expressive Platforms

Our findings extend Lai and Tsai's blueprint of the Pepper robot in catering services [17], which emphasized functional tasks such as welcoming or order-taking. Our research investigated how expressive cues, including gaze direction, movements, and tone of voice, influence first impressions and perceived warmth. Relative to earlier catering studies using less expressive platforms such as Pepper, Furhat's expressivity allowed richer greetings and confirmations, shifting the space of possible interactions. However, our findings indicate that acceptance and competence-related judgments still depend on clear role allocation, exit cues, and staff handover. This supports treating robots as co-performers in hybrid encounters, not replacements for human service [21, 30].

4.5 Limitations and Future Work

Our studies involved modest sized samples with convenience recruitment. In the restaurant environment, noise and crowding affected speech recognition, and novelty may have influenced perceptions since many customers had limited prior exposure to robots. These factors limit the generalizability of our findings but highlight the value of in-the-wild studies for identifying real world challenges. As future work we plan to compare variants of greeting and exit strategies, and examine hybrid service scenarios where roles shift dynamically between staff and robots.

Although we used validated tools such as RoSAS and UMUX-Lite, emotional reactions were captured primarily through self-report and open coding, which may not capture the full nuance or intensity of affective responses. Future work could integrate more physiological or behavioral measures to triangulate emotional data.

5 Conclusion

This paper examined humanoid robots in kiosk-style restaurant encounters through an in-the-wild deployment. Using a service design perspective, we showed how acceptance, perceived competence, discomfort, and social-emotional interaction are shaped by the robot's expressive capabilities and the surrounding servicescape. The field study revealed that brief real-world encounters often evoked mixed affective responses, including curiosity, amusement, and moments of discomfort.

Our findings extend prior work by showing that in short transactional settings, evaluations of usability and competence are formed very quickly, and that social cues such as greetings, gaze, and clear exit signals play a central role in shaping acceptance. By translating these findings into a service blueprint, we contribute an analytical framework for examining and designing robot-mediated service encounters. This demonstrates the value of a service design approach for integrating robots into customer journeys while accounting for both efficiency and experiential qualities.

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